Randomized and low-rank approximation

Constant-loss nuclear-error transfer without matrix ordering

Difficulty: challenging **Importance:** interesting to the community

Status: Partially resolved

Rating rationale: Challenging because a dimension-independent loss must survive removal of matrix ordering; community impact is transferring nuclear-error guarantees between matrix functions.
Topic: Low-rank approximation of matrix functions.

Does a universal constant $C \geq 1$ exist with the following property? For every $n \geq 2$, $1 \leq k < n$, real symmetric positive semidefinite matrices $A, \widehat{A} \in \mathbb{R}^{n\times n}$, $\varepsilon \geq 0$, and continuous operator-monotone function $f : [0,\infty) \to [0,\infty)$,

$$\|A - \widehat{A}_k\|_* \leq (1+\varepsilon)\|A - A_k\|_*$$

implies

$$\|f(A) - f(\widehat{A})_k\|_* \leq (1+C\varepsilon)\|f(A) - f(A)_k\|_*?$$

Here $\|M\|_* = \sum_i \sigma_i(M)$ is the nuclear norm. Operator monotonicity means $f(H) \succeq f(G)$ for every size and every real symmetric $H \succeq G \succeq 0$; functions of matrices are defined spectrally. No ordering between A and $\widehat{A}$ is assumed.

For an ordered orthonormal eigendecomposition $X = \sum_{i=1}^{n} \lambda_i q_i q_i^T$ of a PSD matrix, set

$$X_k = \sum_{i=1}^{k} \lambda_i q_i q_i^T, \qquad f(X)_k = \sum_{i=1}^{k} f(\lambda_i) q_i q_i^T.$$

The assertion must hold for every choice of ordered eigenvectors, using the same choice in both truncations. C must be independent of all inputs, including f.

This is the nuclear-norm instance of the authors' explicit question about a fixed loss in approximation accuracy. It asks whether the ordering condition in existing transfer results can be removed at a controlled price.

References

1. D. Persson, R. A. Meyer, and C. Musco, Algorithm-agnostic low-rank approximation of operator monotone matrix functions, *SIAM Journal on Matrix Analysis and Applications* 46 (2025), 1–21. §1.2, paragraph immediately preceding Table 1; §5, Example 5.3 and closing paragraph; Theorem 2.4 is the ordered predecessor. Equation (2) specifies truncation.
2. D. Persson, T. Chen, and C. Musco, Randomized block-Krylov subspace methods for low-rank approximation of matrix functions, *Linear Algebra and its Applications* 741 (2026), 32–65, abstract and algorithmic scope.

Status check — 2026-09-08

Checked the source's latest listed arXiv v2 (2024-07-04), its 2025 journal record, and the exact counterexample and constant-loss discussion. Searches combining the title, "constant", "counterexamples", "nuclear", "funNyström", and 2026 found no general resolution. Example 5.3 rules out $C = 1$, while the source expressly leaves a larger fixed constant possible. The 2026 Krylov work addresses specific algorithms, not arbitrary PSD approximation pairs. This entry is explicitly a specialization of the source's broader question; the source does not number it as a separate conjecture. No later explicit reaffirmation was located.

Audit — 2026-09-10

Rechecked Theorem 2.4 and Example 5.3: ordered pairs satisfy $C = 1$, while unordered pairs can violate that constant. A larger universal constant remains open. Constant-loss nuclear-transfer searches found no resolution; the partial tag refers only to the ordered subclass.